INNOVATOR TEST 2
MATH MODULE 1

1. A group of friends is eating at a diner. Each entree costs $\$5.25$, and each soda costs $\$1.50$. If d people order an entree and p people order a soda, which of the following expressions represents the total number of dollars paid by the group?

NEXT QUESTION

- A $5.25p + 1.5d$
- B $6.75dp$
- C $5.25d + 1.5p$
- D $5.25d - 1.5p$

2. $$y > -3x + 5$$
$$y > 2x?$$

Which ordered pair (x, y) is a solution to the given system of inequalities?

PREV. QUESTION NEXT QUESTION

- A $(1, 4)$
- B $(2, 1)$
- C $(2, 4)$
- D $(4, 3)$

3. If $3x - y = 11$ and $2x - 2y = 2$, what is the value of $x + y$?

PREV. QUESTION | NEXT QUESTION

A 3

B 5

C 9

D 13

4. $$(t - 5) + 2(s - 3) = 8$$
$$2(t - 5) - 3(s - 3) = -19$$

The solution to the given system of equations is (t, s). What is the value of s?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

5. Which of the following expressions is equivalent to $(x - 2)^2(x + 2)^2$?

PREV. QUESTION | NEXT QUESTION

A $x^4 - 8x^2 + 16$

B $x^4 - 16$

C $x^4 + 8x + 16$

D $x^4 + 16$

6. If a is a solution to the equation $|2x - 3| = 11$ and $a > 0$, what is the value of a?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

7. What is the value when 80 is increased by 200%?

PREV. QUESTION | NEXT QUESTION

A 160

B 200

C 240

D 280

8. The radius of a right cylinder is 6 inches, and the length of its height is twice the length of the radius. What is the volume of the cylinder, in cubic inches?

PREV. QUESTION | NEXT QUESTION

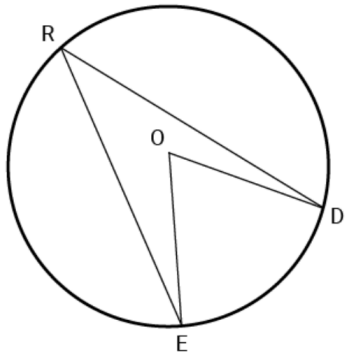
A 108π

B 216π

C 432π

D 864π

9.



The circle shown has center O. If the measure of angle DRE is 35°, what is the measure of angle EOD?

PREV. QUESTION | NEXT QUESTION

× A 35°

× B 55°

× C 60°

× D 70°

COME BACK LATER

10. $5(x+3)-3(2-x)=px+7$

In the given equation, p is a constant. If the equation has no solution, what is the value of p ?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

11. $4cx-4=x-4$

In the given equation, c is a constant. If the equation has infinitely many solutions, what is the value of c?

PREV. QUESTION | NEXT QUESTION

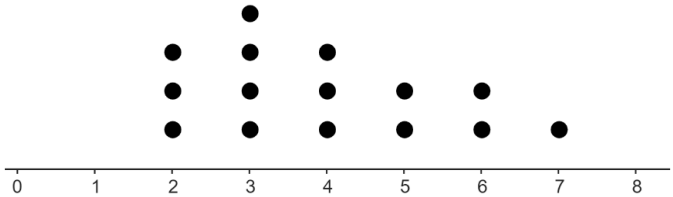
× A 2

× B 1

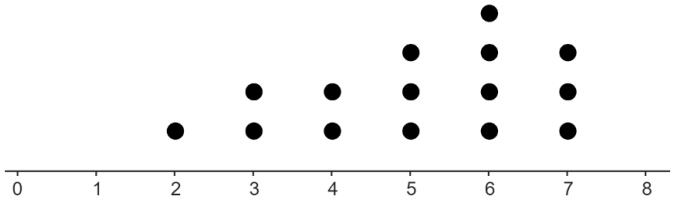
× C $\frac{1}{2}$

× D $\frac{1}{4}$

12



SET 1



SET 2

The dot plots shown summarize the data in two sets, Set 1 and Set 2. Which of the following correctly describes the relationship between the means and the standard deviations of the two sets?

PREV. QUESTION | NEXT QUESTION

A. The standard deviation of Set 1 is greater than the standard deviation of Set 2, and the mean of Set 1 is less than the mean of Set 2.

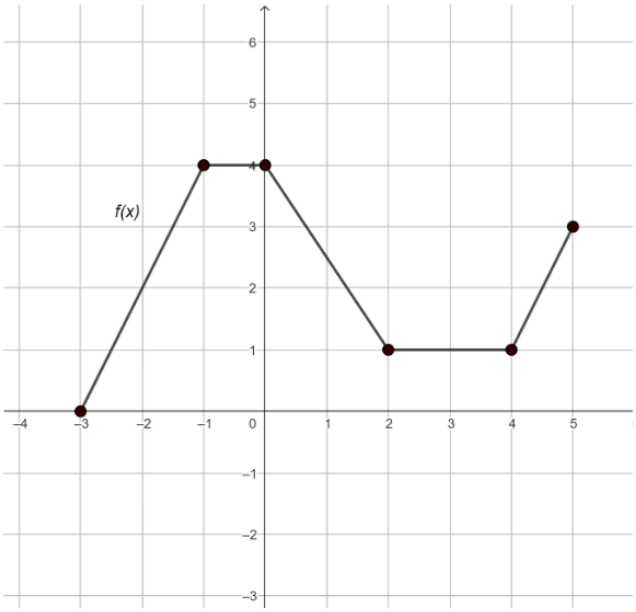
B. The standard deviation of Set 1 is less than the standard deviation of Set 2, and the mean of Set 1 is greater than the mean of Set 2.

C. The standard deviation of the two sets is equal, and the mean of Set 1 is less than the mean of Set 2.

D. The standard deviation of the two sets is equal, and the mean of Set 1 is greater than the mean of Set 2.

COME BACK LATER

13.



PREV. QUESTION | NEXT QUESTION

A 0

B 1

C 2

D 3

COME BACK LATER

The complete graph of the function f in the xy-plane is shown. For how many values of x is $f(x)$ equal to 2.5 ?

14. $(5x^4 + x^3) - (3x^3 + 4x^2) - (x^2 - 8x)$

The given expression can be written in the form $ax^4 + bx^3 + cx^2 + dx$, where a, b, c, and d are constants. What is value of $c + d$?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

15. $f(x) = 3x^2 + 13x - 30$

The given equation can be written in the form $f(x) = (3x - a)(x + b)$. What is the sum of a and b?

PREV. QUESTION | NEXT QUESTION

- A 27
- B 13
- C 11
- D 1

16. $$V(t) = 25(0.88)^{\frac{t}{3}}$$

The given equation models the value that remains, $V(t)$, in millions of dollars, of a new piece of equipment after t years. What is the meaning of 25 in this context?

PREV. QUESTION | NEXT QUESTION

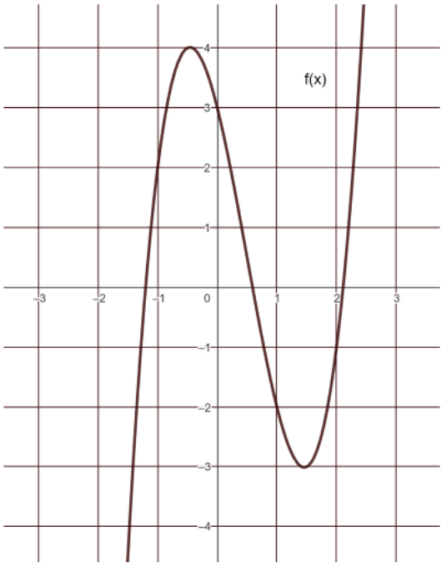
- A The number of years until the piece of equipment has no value.
- B The value of the equipment, in millions of dollars, after the first year.
- C The number of years until the company expects to replace this piece of equipment.
- D The value of equipment, in millions of dollars, when new.

17. In 2008, 20 invasive water flowers were found in a waterway in Virginia. The number of invasive flowers increases by 125% every year. Which of the following equations best describes the number of flowers, $H(t)$, estimated in the waterway t years after 2008?

PREV. QUESTION | NEXT QUESTION

- A $H(t) = 20(2.25)^t$
- B $H(t) = 20(1.25)^t$
- C $H(t) = 1.25(20)^t$
- D $H(t) = 1.25(1.20)^t$

18.



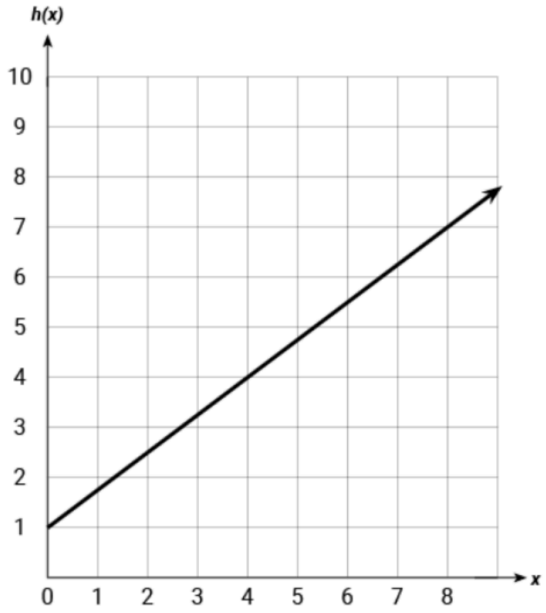
The graph of $f(x)$ is shown. If the function $g(x) = f(x) + 2$, what must be true of $g(x)$?

PREV. QUESTION | NEXT QUESTION

- A $g(0) = 1$ and $g(1) = -4$
- B $g(-2) = 3$ and $g(-1) = -2$
- C $g(2) = 3$ and $g(3) = -2$
- D $g(0) = 5$ and $g(1) = 0$

COME BACK LATER

19



The graph of the function $h(x)$ shows how high, in meters, a balloon is in the air after x seconds. What is the best interpretation of the function $h(x) + 3$?

PREV. QUESTION | NEXT QUESTION

- A The balloon is released at the same height and the height of the balloon is tripled each second.
- B The balloon is released at a height of 3 meters and the height of the balloon increases at the same rate.
- C The balloon is released 3 seconds later and the height of the balloon increases by 3 meters each second.
- D The balloon is released at a height of 4 meters and the height of the balloon increases at the same rate.

COME BACK LATER

20. Circle P has a radius of $5a$ and circle Q has a radius of $155a$, where a is a positive constant. The area of circle Q is how many times the area of circle P?

PREV. QUESTION | NEXT QUESTION

- A 31
- B 62
- C 155
- D 961

21. ABC is a right triangle, where angle B is 90° and angle C is 30°. If $AC = 32$ inches, what is the area of triangle ABC, in inches?

PREV. QUESTION | NEXT QUESTION

× A $16\sqrt{2}$

× B $16\sqrt{3}$

× C $128\sqrt{2}$

× D $128\sqrt{3}$

22. The table shows the average weekly bread sales at Max's Bakery.

	Sliced	Unsliced	Total
Rye	54	41	95
Whole Grain	119	84	203
Sourdough	32	55	87
Total	205	180	385

If a customer who purchased unsliced bread is selected at random, what is the probability that the customer has purchased whole grain bread?

PREV. QUESTION

× A $\frac{84}{203}$

× B $\frac{203}{385}$

× C $\frac{84}{180}$

× D $\frac{180}{203}$

MATH MODULE 2B

1. Maximum Mail Order Company upgrades its computers each year. The function $P(t) = 36t + 410$ models the company's budget for upgrades, in thousands of dollars, t years after 2004. What is the best interpretation of 36 in this context?

PREV. QUESTION | NEXT QUESTION

- A. The number of years until the budget is completely spent
- B. The amount budgeted, in thousands of dollars, in the year 2004
- C. The annual increase, in thousands of dollars, in the budget each year after 2004
- D. The number of years in which the budget will increase after 2004

2. Which expression is equivalent to $\frac{4x^2-3}{2x+\sqrt{3}}$?

PREV. QUESTION | NEXT QUESTION

- A. $2x - 3$
- B. $2x - \sqrt{3}$
- C. $2x + \sqrt{3}$
- D. $2x + 3$

3. $$y - 7 = 3^x - 5$$

What is the y–intercept for the given equation?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

4. A linear model estimates that the mean weight of a calf in New Zealand, at birth, is 34 kilograms. At 180 days after birth, the mean weight is 272 kilograms. Which function best defines this model, where $f(x)$ is the mean weight of a calf, in kilograms, x days after it is born, for $0 \leq x \leq 180$?

PREV. QUESTION | NEXT QUESTION

- A $f(x) = 0.76x + 34$
- B $f(x) = 0.76x + 180$
- C $f(x) = 1.32x + 34$
- D $f(x) = 34x + 272$

5.

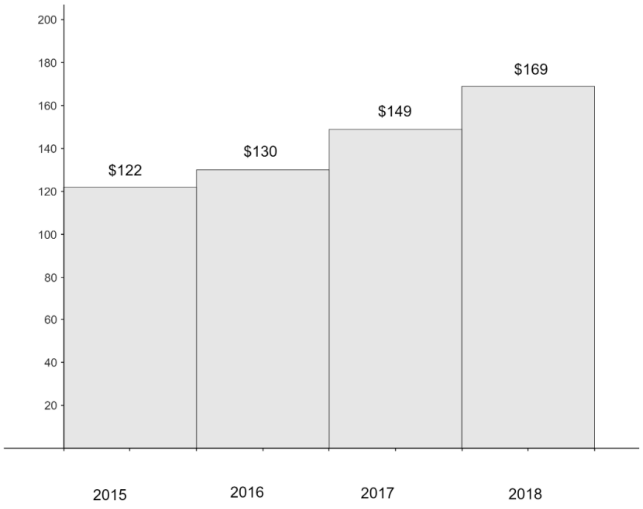
x	$f(x)$
1	3
3	6
5	9
7	12
9	15

For the function f, the table shows some values of x and their corresponding values of $f(x)$. If $f(1) = t$, what is the value of $f(3t)$?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

6.



The chart shows the average cost of housing, in dollars per square foot, in Nebraska from 2015 to 2018. By approximately what percent did the average cost of housing increase over this period?

PREV. QUESTION | NEXT QUESTION

- A 20%
- B 28%
- C 39%
- D 47%

COME BACK LATER

7.

$$p(t) = 2{,}034(3)^{\frac{t}{14}}$$

The given function shows the number of bacteria in a pond after t days since the bacteria began to populate the pond. Which of the following is the best interpretation of $(3)^{\frac{t}{14}}$ in the equation?

PREV. QUESTION | NEXT QUESTION

- A The number of bacteria at the beginning of the study
- B The number of bacteria triples every two weeks
- C The number of bacteria increases by 3 every two weeks
- D The number of bacteria in the pond after two weeks

8.

$$12x^4 - 8x^3 - 15x^2$$

The given expression is equal to $x^2(qx + 5)(rx - 3)$, where $q > 0$ and $r > 0$. What is the value of $q + r$?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

9. Line a is parallel to line b. Line a goes through the points $(-4, 0)$ and $(0, 6)$. If line b goes through the point $(-6, 0)$, what is the equation for line b?

PREV. QUESTION | NEXT QUESTION

- A $y = \frac{2}{3}x - 6$
- B $y = \frac{2}{3}x + 4$
- C $y = \frac{3}{2}x + 6$
- D $y = \frac{3}{2}x + 9$

10. The exterior angle of an equilateral triangle is equal to $(8x - 20)^\circ$. What is the value of x?

PREV. QUESTION | NEXT QUESTION

- A 5
- B 10
- C 12.5
- D 17.5

11. $(x-1)(x+2)=x(x-3)$

How many solutions does the given equation have?

PREV. QUESTION | NEXT QUESTION

- A 0
- B 1
- C 2
- D 4

12. $y>-2x-1$
$3y<x+9$

Which of the following coordinates would be true for the given system of inequalities?

PREV. QUESTION | NEXT QUESTION

- A $(-3,3)$
- B $(-2,1)$
- C $(1,4)$
- D $(3,1)$

13. $9x-14y=-3$
$ax-by=6$

If the given system of equations has infinite solutions, what is the value of b?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

14.

$$f(x) = 6x^2 - 45x + 75$$

Which of the following equivalent forms of the given function f displays the x-intercepts as constants or coefficients?

PREV. QUESTION | NEXT QUESTION

× A $f(x) = (2x - 5)(3x - 15)$

× B $f(x) = 3x(2x - 15) + 75$

× C $f(x) = 6(x - \frac{15}{4})^2 + \frac{75}{8}$

× D $f(x) = 6(x - 5)(x - \frac{5}{2})$

15.

$$y = 4$$
$$y = x^2 + 12x + n$$

In the given system of equations, n is a positive constant. For what value of n does the system have exactly one real solution?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

16.

$$y \geq 2(x-3)^2 + 5$$
$$y \leq x + 3$$

If (a, b) is a solution to the given system of inequalities, what is the minimum value of b?

PREV. QUESTION | NEXT QUESTION

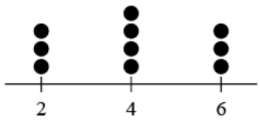
- A -5
- B -3
- C 3
- D 5

17. Biology students grew a culture of bacteria at a constant temperature of $55^\circ C$ for 3 days. The students began with 200 spores and the number of spores doubled every 12 hours. Of the equations given, which can be used to correctly calculate the number of spores in the bacteria culture at the end of 3 days?

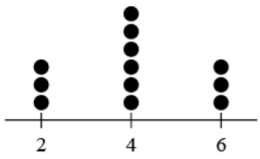
PREV. QUESTION | NEXT QUESTION

- A $y = 200(12)^3$
- B $y = 200(2)^3$
- C $y = 200(2)^6$
- D $y = 200(12)^2$

18.



DATA SET J



DATA SET K

Which of the statements best compares the means and standard deviations of Data Set J and Data Set K?

PREV. QUESTION | NEXT QUESTION

× A The mean of Data Set J is less than the mean of Data Set K, and the standard deviations of the two sets are equal.

× B The mean of Data Set J is greater than the mean of Data Set K, and the standard deviations of the two sets are equal.

× C The means of the two sets are equal, and the standard deviation of Data Set J is less than the standard deviation of Data Set K.

× D The means of the two sets are equal, and the standard deviation of Data Set J is greater than the standard deviation of Data Set K.

19. What is the circumference of the circle with an equation of $x^2 + 6x + y^2 - 4y = 51$?

PREV. QUESTION | NEXT QUESTION

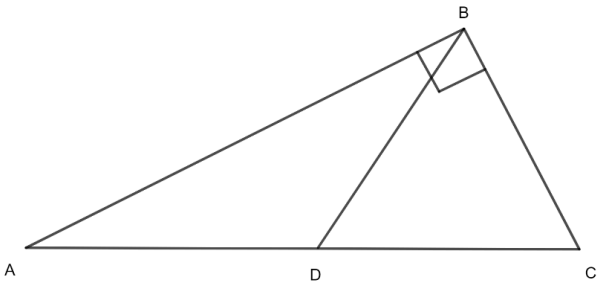
× A 8π

× B 14.28π

× C 16π

× D 64π

20.



In the triangle shown, $\angle ABC = 90^\circ$. Line segment BD is shown with point D lying on line AC. What is the value of $\sin \angle ABD - \cos \angle DBC$?

PREV. QUESTION | NEXT QUESTION

- A 0
- B 30
- C 45
- D 90

COME BACK LATER

21.

$$y = (x - 6)(x + 6)$$
$$y = k$$

In the system of equations shown, k is an integer constant. What is the largest value of k for which the system has no solution?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

22. The base of the exponential function $g(x)$ is k, where k is a positive real number, and k is equivalent to the value of the y–coordinate of the y–intercept of the graph of $g(x)$. Which of the following functions describes the exponential function $g(x-1)$?

PREV. QUESTION | NEXT QUESTION

A $g(x-1)=k^{x}$

B $g(x-1)=k^{x+1}$

C $g(x-1)=k^{x}+k$

D $g(x-1)=k^{x-1}+k$